$L_{grid}\left(i_{cc}\right)$

$L_{\sigma}^{tr}\left(u_{cc},P_{nom}^{tr}\right)$

C_{Fi}

R_{brake}

L_{Fi}

$u_{s}\left(t\right)$

$u_{g}\left(t\right)$

$u_{dc}\left(t\right)$

$i_{s}\left(t\right)$

$i_{g}\left(t\right)$

$n\cdot C_{Fu}$

R_{Fu}/n

L_{Fu}/n

$n\cdot i_{s}\left(t\right)$

$n\cdot i_{g}\left(t\right)$

L_{g}

n:

$e\left(t\right)=\omega\left(t\right)\psi^{m}$

$\frac{d\vartheta}{dt}=\omega$

$$L_{s}\left(\vartheta\right)=\begin{bmatrix} l_{\mu}+l_{a}-l_{b}\cos\left(2\vartheta\right) & -\frac{1}{2}l_{a}-l_{b}\cos\left(2\left(\vartheta-\frac{\pi}{3}\right)\right) & -\frac{1}{2}l_{a}-l_{b}\cos\left(2\left(\vartheta+\frac{\pi}{3}\right)\right) \\ -\frac{1}{2}l_{a}-l_{b}\cos\left(2\left(\vartheta-\frac{\pi}{3}\right)\right) & l_{\mu}+l_{a}-l_{b}\cos\left(2\left(\vartheta-\frac{2\pi}{3}\right)\right) & -\frac{1}{2}l_{a}-l_{b}\cos\left(2\left(\vartheta+\pi\right)\right) \\ -\frac{1}{2}l_{a}-l_{b}\cos\left(2\left(\vartheta+\frac{\pi}{3}\right)\right) & -\frac{1}{2}l_{a}-l_{b}\cos\left(2\left(\vartheta+\pi\right)\right) & l_{\mu}+l_{a}-l_{b}\cos\left(2\left(\vartheta+\frac{2\pi}{3}\right)\right) \end{bmatrix}$$

$e_{1}\left(t\right)$

$e_{2}\left(t\right)$

$e_{3}\left(t\right)$

$$\begin{cases} \frac{d\psi_{r}^{u}}{dt}=-\omega\left[\psi^{m}\sin\left(\vartheta\right)-5h_{5}\psi^{m}\sin\left(-5\vartheta\right)\right] \\ \frac{d\psi_{r}^{v}}{dt}=-\omega\left[\psi^{m}\sin\left(\vartheta-\frac{2\pi}{3}\right)-5h_{5}\psi^{m}\sin\left(-5\vartheta-\frac{2\pi}{3}\right)\right] \\ \frac{d\psi_{r}^{w}}{dt}=-\omega\left[\psi^{m}\sin\left(\vartheta+\frac{2\pi}{3}\right)-5h_{5}\psi^{m}\sin\left(-5\vartheta+\frac{2\pi}{3}\right)\right] \end{cases}$$

rst

$\alpha\beta$

$\xi\eta_{p,n}$

$\xi\eta_{n}$

$\xi\eta$

$-\breve{\vartheta}\left(t\right)$

$u_{g}^{u}\left(t\right)$

$u_{g}^{v}\left(t\right)$

$u_{g}^{w}\left(t\right)$

$u_{g}^{\xi}\left(t\right)$

$u_{g}^{\eta}\left(t\right)$

$-\breve{\vartheta}\left(t\right)$

$u_{p}^{\xi}\left(t\right)$

$u_{p}^{\eta}\left(t\right)$

$u_{n}^{\xi}\left(t\right)$

$u_{n}^{\eta}\left(t\right)$

$u_{p}^{\xi}\big|_{m}\left(t\right)$

$u_{p}^{\eta}\big|_{m}\left(t\right)$

$u_{n}^{\xi}\big|_{m}\left(t\right)$

$u_{n}^{\eta}\big|_{m}\left(t\right)$

$x_{m}\left(k\right)=\frac{1}{N}\sum_{i=0}^{N-1}x\left(k-i\right)$

$i_{g}^{u}\left(t\right)$

$i_{g}^{v}\left(t\right)$

$i_{g}^{w}\left(t\right)$

$i_{p}^{\xi}\left(t\right)$

$i_{p}^{\eta}\left(t\right)$

$i_{n}^{\xi}\left(t\right)$

$i_{n}^{\eta}\left(t\right)$

$i_{p}^{\xi}\big|_{m}\left(t\right)$

$i_{p}^{\eta}\big|_{m}\left(t\right)$

$i_{n}^{\xi}\big|_{m}\left(t\right)$

$i_{n}^{\eta}\big|_{m}\left(t\right)$

$i_{p}^{\eta}\big|_{m}^{ref}\left(t\right)$

$i_{n}^{\xi}\big|_{m}^{ref}\left(t\right)$

$i_{n}^{\eta}\big|_{m}^{ref}\left(t\right)$

$i_{p}^{\xi}\big|_{s}^{ref}\left(t\right)$

$i_{p}^{\eta}\big|_{s}^{ref}\left(t\right)$

$i_{n}^{\xi}\big|_{s}^{ref}\left(t\right)$

$i_{n}^{\eta}\big|_{s}^{ref}\left(t\right)$

$i_{su}^{ref}\left(t\right)$

$i_{sv}^{ref}\left(t\right)$

$i_{sw}^{ref}\left(t\right)$

$k_{p}^{vc}+\frac{k_{i}^{vc}}{s}$

$k_{p}^{rc,pos}+\frac{k_{i}^{rc,pos}}{s}$

$k_{p}^{rc,neg}+\frac{k_{i}^{rc,neg}}{s}$

$k_{p}^{cc}+k_{i}^{cc}\frac{s}{s^{2}+2\delta\omega_{res}s+\omega_{res}^{2}}$

$k_{p}+\frac{k_{i}}{s}$

$u_{dc}^{ref}\left(t\right)$

$u_{dc}\left(t\right)$

$i_{p}^{\xi}\big|_{s}^{ref}\left(t\right)$

$i_{s}^{u}\left(t\right)$

$i_{s}^{v}\left(t\right)$

$i_{s}^{w}\left(t\right)$

$u_{ctrl}^{u}\left(t\right)$

$u_{ctrl}^{v}\left(t\right)$

$u_{ctrl}^{w}\left(t\right)$

$u_{ctrl}^{\alpha}\left(t\right)$

$u_{ctrl}^{\beta}\left(t\right)$

$\underset{\sim}{i}_{g}^{uvw}\left(t\right)$

$\underset{\sim}{i}_{s}^{uvw}\left(t\right)$

$\underset{\sim}{u}_{g}^{uvw}\left(t\right)$

$\underset{\sim}{u}_{s}^{uvw}\left(t\right)$

$d_{u}^{ctrl}\left(t\right)$

$d_{v}^{ctrl}\left(t\right)$

$d_{w}^{ctrl}\left(t\right)$